

M3: Exam 1 -- covers modules 1 and 2

Results for Jonathan Elder

Score for this quiz: **168.89** out of 200

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This attempt took 52 minutes.

Question 1

8 / 8 pts

The following sentence is a proposition

"7 is an even number."

☒ True

☐ False

Question 2

8 / 8 pts

The following sentence is a proposition

"Did you do the homework?"

☐ True

☒ False

Question 3

4 / 4 pts

Assuming p is false, q is false and z is true.

What is the boolean value of the expression:

$\neg z \vee q \wedge p$

☒ False

☐ True

Question 4

4 / 4 pts

Complete the following truth table (note: this table does not consider all possible values of the variables, just the ones of interest in this problem).

a	b	c	$a \vee (b \wedge \neg c)$
T	T	T	[Select] ▼
T	F	F	T
F	F	T	[Select] ▼

Answer 1:

T

Answer 2:

T

Answer 3:

F

Question 5

4 / 4 pts

Assume p is true, q and r are false. What is the value of:

$$\neg(p \vee q) \wedge (\neg p \vee r)$$

☐ True

☒ False

Question 6

4 / 4 pts

Which of the following is the **contrapositive** of:

"If I practice this piece, I will perform it correctly."

☒ If I did not perform this piece correctly then I did not practice it.

☐ Practice makes perfect.

☐ If I don't practice this piece then I will not perform it correctly.

☐ Practice makes better.

☐ If I perform this piece correctly then I practiced it.

Question 7

4 / 4 pts

Which of the following is the **inverse** of

"If Little Debbie would stop making Swiss Cake Rolls, I would loose 30 pounds"

- ☐ Little Debbie needs to stop making Swiss Cake Rolls before someone dies of heart failure
- ☐ Ummm, Swiss Cake Rolls.
- ☐ If I do not loose 30 pounds then Little Debbie did not stop making Swiss Cake Rolls.
- ☒ If Little Debbie does not stop making Swiss Cake Rolls, I will not loose 30 pounds.
- ☐ If I loose 30 pounds then Little Debbie stopped making Swiss Cake Rolls.

Question 8

4 / 4 pts

Which of the following is the **converse** of:

"If I explained the problem, she understood it."

- ☒ If she understood the problem then I explained it.
- ☐ If she didn't understand the problem then I didn't explain it.
- ☐ If I didn't explain the problem, she didn't understand it.
- ☐ I understand everything that she explains to me
- ☐ She understands everything I tell her.

Question 9

4 / 4 pts

What is the **converse** of:

$$q \rightarrow (r \vee s)$$

- ☐ $\neg q \rightarrow (\neg r \wedge \neg s)$
- ☐ $q \rightarrow \neg(r \vee s)$
- ☐ $(\neg r \wedge \neg s) \rightarrow \neg q$
- ☒ $(r \vee s) \rightarrow q$

Question 10

4 / 4 pts

What is the **contrapositive** of:

$$(r \wedge s) \rightarrow p$$

- ☐ $(r \wedge s) \rightarrow p$
- ☐ $\neg(r \wedge s) \rightarrow \neg p$
- ☒ $\neg p \rightarrow \neg(r \wedge s)$
- ☐ $(r \wedge s) \rightarrow \neg p$

Question 11

0 / 8 pts

If the proposition

$$p \rightarrow \neg q$$

is true, which of the following **must also be true**? Select all that apply.

- ☒ $\neg q \rightarrow \neg p$
- ☒ $q \Rightarrow p$
- ☐ $q \rightarrow p$
- ☐ $q \rightarrow \neg p$

Question 12

2.67 / 8 pts

Select all of the following statements that are **logically equivalent** to:

$$x \rightarrow (y \vee z)$$

- ☒ $(\neg y \wedge \neg z) \rightarrow \neg x$
- ☒ $\neg(y \vee z) \rightarrow \neg x$
- ☒ $\neg x \rightarrow \neg(y \vee z)$
- ☐ $\neg x \vee y \vee z$
- ☐ $x \wedge (y \vee z)$

Question 13

8 / 8 pts

In the following proof, indicate which law of propositional logic was used to reach the indicated step. For example, on step 2 you would indicate what law was used to get from step 1 to step 2.

Prove the following logical equivalence:

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

1. $(p \rightarrow q) \wedge (p \rightarrow r)$

2. $(\neg p \vee q) \wedge (p \rightarrow r)$ [Select]

3. $(\neg p \vee q) \wedge (\neg p \vee r)$ [Select]

4. $\neg p \vee (q \wedge r)$ [Select]

5. $p \rightarrow (q \wedge r)$ [Select]

Be sure to select an answer for each dropdown before you submit your solution!

Answer 1:

Conditional identity

Answer 2:

Conditional identity

Answer 3:

Distributive

Answer 4:

Conditional identity

Question 14

8 / 8 pts

In the following proof, indicate which law of propositional logic was used to reach the indicated step. For example, on step 4 you would indicate what law was used to get from step 3 to step 4.

Prove the following logical equivalence:

$$p \leftrightarrow (p \wedge r) \equiv \neg p \vee r$$

1. $p \leftrightarrow (p \wedge r)$

2. $(p \rightarrow (p \wedge r)) \wedge ((p \wedge r) \rightarrow p)$

3. $(p \rightarrow (p \wedge r)) \wedge (\neg(p \wedge r) \vee p)$

4. $(p \rightarrow (p \wedge r)) \wedge ((\neg p \vee \neg r) \vee p)$ [Select]

5. $(p \rightarrow (p \wedge r)) \wedge ((\neg r \vee \neg p) \vee p)$ Commutative

6. $(p \rightarrow (p \wedge r)) \wedge (\neg r \vee (\neg p \vee p))$ [Select]

7. $(p \rightarrow (p \wedge r)) \wedge (\neg r \vee T)$ [Select]

8. $(p \rightarrow (p \wedge r)) \wedge T$ [Select]

9. $p \rightarrow (p \wedge r)$ [Select]

10. $\neg p \vee (p \wedge r)$

11. $(\neg p \vee p) \wedge (\neg p \vee r)$

12. $T \wedge (\neg p \vee r)$

13. $(\neg p \vee r) \wedge T$ Commutative

14. $\neg p \vee r$ Identity

Be sure to select an answer for each dropdown before you submit your solution!

Answer 1:

DeMorgan's

Answer 2:

Commutative

Answer 3:

Associative

Answer 4:

Complement

Answer 5:

Domination

Answer 6:

Identity

Answer 7:

Commutative

Answer 8:

Identity

Question 15

8 / 8 pts

Which of the following is the correct translation of the phrase

"There is a real number whose cube is less than its square."

to mathematical notation?

☐ $x^3 < x^2$

☒ $\exists x(x^3 < x^2)$

☐ $\forall x(3x < 2x)$

☐ $\forall x(x^3 < x^2)$

Question 16

4 / 4 pts

Which of the following is the correct translation of

"Everyone who eats Doritos becomes addicted to them"

where

$E(x)$ = x eats Doritos

$A(x)$ = x becomes addicted to them.

☒ $\forall x(E(x) \rightarrow A(x))$

☐ $\exists x(E(x) \rightarrow A(x))$

☐ $\forall x(E(x) \vee A(x))$

☐ $\forall x(E(x) \wedge A(x))$

Question 17

4 / 4 pts

Which of the following is the correct translation of

"Someone who bought the fruit did not like it"

where

$B(x)$ x bought the fruit

$L(x)$ x **liked** the fruit

☐ $\exists x(B(x) \rightarrow L(x))$

☐ $\forall x(B(x) \wedge \neg L(x))$

☒ $\exists x(B(x) \wedge \neg L(x))$

☐ $\exists x(B(x) \wedge L(x))$

☐ $\forall x(B(x) \rightarrow L(x))$

Question 18

8 / 8 pts

Given the following predicates:

$Z(x)$: x is a zebra

$S(x)$: x has stripes

convert the following to English:

$$\exists x(Z(x) \wedge \neg S(x))$$

☐ If an animal is a zebra it does not have stripes

☐ All zebras do not have stripes

☐ There might be a zebra that does not have stripes.

☒ There is a zebra that does not have stripes.

Question 19

4 / 8 pts

The domain of discourse for this problem is the integers 1, 2, 3 and 4. The table below shows the value of three predicates for each of the elements in the domain. For example, $Q(1)$ is true because the truth value in the row labeled "1", column labeled "Q" is T.

x	$P(x)$	$Q(x)$	$R(x)$
1	F	T	T
2	F	T	T
3	F	T	F
4	T	T	F

Select all **true** statements from the list below.

- ☒ $\exists x P(x)$
- ☐ $\forall x P(x)$
- ☒ $\exists x (P(x) \wedge R(x))$
- ☐ $\forall x (\neg P(x))$
- ☒ $\forall x (Q(x) \vee R(x))$
- ☐ $\forall x R(x)$
- ☒ $\exists x R(x)$
- ☐ $\exists y Q(y)$
- ☐ $\exists x \neg Q(x)$

Question 20

0 / 8 pts

Which of the following is the **logical negation** of the following phrase:

"Some people have glasses but cannot read."

- ☐ Some people with glasses can read.
- ☐ Glasses help make people read better.
- ☒ There is a person with glasses who cannot read.
- ☐ All people either don't have glasses or can read (or both)

Question 21

4 / 8 pts

Select each **valid** argument from the list below.

☒
$$\frac{p \rightarrow q}{p} \therefore q$$

☒
$$\frac{p \rightarrow q}{q} \therefore p$$

☐
$$\frac{p \vee q}{\neg q} \therefore q$$

☒
$$\frac{p \vee q}{\neg p} \therefore q$$

☐
$$\frac{p \rightarrow q}{\neg p} \therefore \neg q$$

Question 22

8 / 8 pts

Select the correct form for the argument.

23 is a prime number

if 23 is prime then 23 is not divisible by 7

Therefore 23 is not divisible by 7

☒
$$\frac{p}{p \rightarrow q} \therefore q$$

☐
$$\frac{p \wedge q}{\neg q} \therefore p$$

☐
$$\frac{p}{p \wedge q} \therefore q$$

☐
$$\frac{p \vee q}{q} \therefore p$$

Question 23

8 / 8 pts

Given the following two statements:

1. $p \rightarrow (q \rightarrow r)$
2. p

What statement would result from the application of Modus ponens to these two statements?

☒ $q \rightarrow r$

☐ $q \vee r$

☐ r

☐ q

Question 24

6.22 / 8 pts

Below is a proof that the following argument is valid:

$$\begin{array}{l} p \rightarrow q \\ r \rightarrow u \\ \hline p \wedge r \\ \hline \therefore q \wedge u \end{array}$$

In each line of the proof, select the rule of inference that was used to arrive at that line. If the line used a Propositional logic law (such as the commutative law, for example) rather than a rule of inference then just select "Basic logic law"

1. $p \rightarrow q$ [Select]
2. $p \wedge r$ [Select]
3. p [Select]
4. q [Select]
5. $r \rightarrow u$ [Select]
6. $r \wedge p$ [Select]
7. r [Select]
8. u [Select]
9. $q \wedge u$ [Select]

Answer 1:

Hypothesis

Answer 2:

Hypothesis

Answer 3:

Simplification

Answer 4:

Modus ponens

Answer 5:

Hypothesis

Answer 6:

Basic logic law

Answer 7:

Simplification

Answer 8:

Basic logic law

Answer 9:

Resolution

Question 25

8 / 8 pts

The following argument is valid:

"All donut shops make glazed donuts. Dunkin' Donuts is a donut shop. Therefore Dunkin' Donuts makes glazed donuts."

☒ True☐ False

Question 26

8 / 8 pts

Finding that a statement holds for one case is sufficient to prove an existential statement.

☒ True☐ False

Question 27

8 / 8 pts

Finding a single example where a statement is false is enough to **disprove** a universally quantified statement.

☒ True☐ False

Question 28

8 / 8 pts

Select all values of x in the list below that act as a counter examples to

$$\forall x (\sqrt{x+1} > x)$$

where the domain of discourse is non-negative integers.

☐ 0☒ 3☐ 1

Question 29

8 / 8 pts

There exists an integer x such that $x^2 - x = 0$

☒ True☐ False

Question 30

8 / 8 pts

For all positive integers n the quantity $n^2 - 1$ is prime.

☐ True☒ False

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